VINUNIVERSITY

Test duration: 1 hour and 30 minutes

Name: **Student ID:**

Midterm Practice Test 3 – Solutions – Fall 2026

COMP3040/5030 Computer Vision Unofficial practice test, prepared by Lê Tăng Phú Quý

Instructions:

- Academic honesty is required in all work you submit to be graded. You should **NOT** copy or share your solutions with other students to avoid plagiarism issues.
- For short answer questions, please make your answers as **concise** as possible.
- This is a closed-book exam and NO Internet access is allowed. You are allowed 1 page of **hand-written** note (both sides).
- Drafting paper and pen/pencil are allowed. We will provide the drafting paper to you.
- Late submission without an approved extension will not be graded at all.
- You must **RETURN** this hard copy of the midterm exam at the end of the exam period.
- During the exam, you are **NOT** allowed to leave the room or/and use the restroom.
- You are required to **NOT** use your mobile phone in the exam room. Please put it on the table or in your bag.
- Good luck!

About the exam: There are **05** problems with **100 points** total, and some have multiple parts. Please check through the pages before you start.

Problem	Content	Point (Undergraduates)	Type
1	Fundamentals of Computer Vision	20	short answer
2	Image Filtering	25	long answer
3	Feature Detection and Matching	25	long answer
4	Fitting and Image Transformations	25	long answer
5	System Design	05	long answer
Total		**100**	

Problem 1. Fundamentals of Computer Vision

Give a short answer for each of the following question:

1.1. A 2 m pole stands 10 m from a camera with a 35 mm focal length. How tall is its image (mm)? At 20 m? Why can a single image not reveal distance? (5 pts)

Solution. $fH/Z = 35 \cdot 2/10 = 7$ mm; at 20 m: 3.5 mm. Every point on the ray $t\mathbf{p}$ projects to the same pixel: a 4 m pole at 20 m looks exactly like a 2 m pole at 10 m.

1.2. `a = np.uint8(200)`. What do `a + np.uint8(100)` and `cv2.add(a, 100)` give? A 2×2 average pooling of 1, 2, 2, 2 stored into a `uint8` array gives what? (5 pts)

Solution. NumPy wraps: $300-256 = 44$; OpenCV saturates: 255. The mean 1.75 is **truncated** when stored as `uint8` $\Rightarrow$ 1 (rounding first would give 2).

1.3. Minimum point correspondences to estimate: a translation, a similarity, an affine map, a homography? State the general rule. (5 pts)

Solution. 1, 2, 3, 4. Each pair gives 2 equations; the DOF are 2, 4, 6, 8 $\Rightarrow \lceil \text{DOF}/2 \rceil$ (a homography has 8 DOF since it is defined up to scale).

1.4. An 8-bit image has an MSE of 25 against the original. Compute the PSNR. How must the noise change to gain 6 dB? (5 pts)

Solution. PSNR $= 10\log_{10}(255^2/25) \approx 34.15$ dB. +6 dB $\Leftrightarrow$ MSE $\div 4$ $\Leftrightarrow$ the noise standard deviation halves (e.g. averaging 4 independent frames).

Problem 2. Image Filtering

2.1. Four pixels $F(0,0) = 10$, $F(1,0) = 20$, $F(0,1) = 30$, $F(1,1) = 50$ (arguments (x, y)). (a) Bilinear interpolation at $(0.5, 0.5)$ and $(0.25, 0.75)$. (b) Upscale $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ ×2 with nearest neighbour. (10 pts)

Solution. (a) $(10+20+30+50)/4 = 27.5$. With $\alpha = 0.25$, $\beta = 0.75$: $0.75 \cdot 0.25 \cdot 10 + 0.25 \cdot 0.25 \cdot 20 + 0.75 \cdot 0.75 \cdot 30 + 0.25 \cdot 0.75 \cdot 50 = 1.875 + 1.25 + 16.875 + 9.375 = 29.375$. (b) Each pixel becomes a 2×2 block: $\begin{bmatrix} 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 2 \\ 3 & 3 & 4 & 4 \\ 3 & 3 & 4 & 4 \end{bmatrix}$.

2.2. Each row of a 7×7 image is $[0, 0, 0, 0, 10, 10, 10]$ (a vertical step edge). Correlate horizontally with $[-1, 0, 1]$. (a) Compute one output row with **zero padding**. (b) What does the value in the last column mean? Redo with **replicate padding**. (c) How many local maxima are in the output of (b)? (15 pts)

Solution. (a) $o[i] = f[i+1] - f[i-1]$: $[0, 0, 0, 10, 10, 0, -10]$. (b) The -10 at the right border is a **spurious edge** created by the zero padding (a $10 \to 0$ step). Replicate: $f[7] = 10 \Rightarrow [0, 0, 0, 10, 10, 0, 0]$. (c) One plateau per row (columns 3–4) $\Rightarrow$ **7** maxima; the edge is 2 px thick because the kernel spans 3 — NMS is needed to thin it.

Problem 3. Feature Detection and Matching

3.1. A Harris window (uniform weights) contains 4 gradients (I_x, I_y): $(10,0), (0,10), (10,10), (-10,10)$. (a) Compute M. (b) With $k = 0.05$ compute R and classify. (c) Redo with $(10,0), (20,0), (10,0), (-10,0)$. (13 pts)

Solution. (a) $\sum I_x^2 = 300$, $\sum I_y^2 = 300$, $\sum I_x I_y = 0 + 0 + 100 - 100 = 0 \Rightarrow M = \mathrm{diag}(300, 300)$. (b) $R = 90\,000 - 0.05 \cdot 600^2 = 72\,000 > 0 \Rightarrow$ **corner** (gradients in several directions). (c) $M = \mathrm{diag}(600, 0)$, $R = 0 - 0.05 \cdot 600^2 = -18\,000 < 0 \Rightarrow$ **edge** (all gradients parallel).

3.2. 1-D template $T = [1,2,3]$, image $I = [0,2,4,6,3,1]$. (a) Compute the SSD map ('valid') and the best position. (b) Compute the ZNCC at positions 0 and 1. (c) Comment on the ZNCC result. (12 pts)

Solution. (a) Patches $[0,2,4], [2,4,6], [4,6,3], [6,3,1] \Rightarrow$ SSD $= 2, 14, 25, 30 \Rightarrow$ best at 0. (b) $T - \bar{T} = [-1,0,1]$; patches 0 and 1 both have $P - \bar{P} \propto [-1,0,1] \Rightarrow$ ZNCC $= 1$ at both. (c) Patch 1 $= 2T$: the same **shape** with a different contrast $\Rightarrow$ ZNCC cannot tell them apart; contrast invariance is not always desirable.

Problem 4. Fitting and Image Transformations

4.1. The estimated inlier ratio is $w = 0.7$ and we want $p = 0.99$. Compute the minimum number of RANSAC iterations for: a line, an affine map, a homography. Comment. (12 pts)

Solution. $N = \lceil \log(1-p) / \log(1-w^s) \rceil$. Line ($s = 2$): $w^2 = 0.49 \Rightarrow 6.8 \to 7$. Affine ($s = 3$): $0.343 \Rightarrow 11.0 \to 11$. Homography ($s = 4$): $0.2401 \Rightarrow 16.8 \to 17$. N grows exponentially with s $\Rightarrow$ always use minimal samples.

4.2. A 400×300 image (width $\times$ height). As in the Lec07 lab: shift the image 100 px right and 100 px down, then rotate $45°$ counter-clockwise about the image centre and scale $\times 2$ (with `cv2.getRotationMatrix2D`). (a) Write both 2×3 matrices and the composed one. (b) Where does pixel $(0,0)$ of the original go? (13 pts)

Solution. (a) $M_1 = \begin{bmatrix} 1 & 0 & 100 \\ 0 & 1 & 100 \end{bmatrix}$; $\alpha = \beta = \sqrt{2} \approx 1.414$, $M_2 = \begin{bmatrix} \alpha & \beta & (1-\alpha)200 - 150\beta \\ -\beta & \alpha & 200\beta + (1-\alpha)150 \end{bmatrix} \approx \begin{bmatrix} 1.414 & 1.414 & -295.0 \\ -1.414 & 1.414 & 220.7 \end{bmatrix}$; composed $\tilde{M} = \tilde{M}_2 \tilde{M}_1$ (M_1 applied first). (b) $(0,0) \to (100,100) \to (-12.13,\ 220.71)$ — off the left edge.

Problem 5. System Design

5. A lab needs to **count and size** nearly circular cells/coins in microscope images (noisy, some touching). Design the algorithm and explain how to choose its parameters. (5 pts)

Solution. Grayscale $\to$ median (impulse noise) or Gaussian $\to$ Canny (σ from the noise level) $\to$ **circle Hough** with a 3-D (a, b, r) accumulator over the known radius range (bounding r saves memory; using the gradient direction lets each edge point vote along one direction) $\to$ peaks + NMS on centre distance (separates touching discs) $\to$ count the peaks, radius = size (px$\to\mu$m via the scale bar). Parameters: vote threshold ~ a fraction of the circumference $2\pi r$,

1 px radius step. (Alternative: 3-point circle RANSAC per connected component.)